

A Trace–SOS Determinantal Inequality for Partitions of the Identity

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Abstract

Let $A_1, \dots, A_m$ be positive semidefinite Hermitian $n \times n$ matrices whose sum is the identity. We prove

$$\det\left(I_n + m(m-1) \sum_{i=1}^m A_i^3\right) \geq (2m-1)^n \det\left(\sum_{i=1}^m A_i^2\right).$$

The argument extends Cubesteak’s trace–sum-of-squares approach to the three-matrix case. Its main ingredient is an explicit family of Gram matrices, valid for every $m \geq 3$, together with an analytic decomposition proving their positive semidefiniteness. In fact, we obtain a stronger quadratic trace inequality and the corresponding inverse-trace bound. The constant is sharp. For positive definite matrices and $m \geq 3$, equality holds precisely when all matrices equal I_n/m ; for $m = 1, 2$, the determinant inequality is an identity. A single optional Python certificate verifies the polynomial trace identity using exact rational arithmetic and an indeterminate parameter.

1 Introduction and main result

Cubesteak proved the following inequality on Mathematics Stack Exchange [1]: if A, B, C are positive definite Hermitian matrices with $A + B + C = I_n$, then

$$\det(I_n + 6(A^3 + B^3 + C^3)) \geq 5^n \det(A^2 + B^2 + C^2).$$

The proof compares determinants through an inverse-trace inequality, removes the inverse by a quadratic variational identity, and establishes positivity by a noncommutative sum-of-squares decomposition. We retain this proof framework and its notation S, T, Φ . The new ingredient is a uniform Gram family that treats an arbitrary number of summands.

Throughout, m, n are positive integers, I_n is the identity in $M_n(\mathbb{C})$, and $A_i = A_i^* \succeq 0$ satisfy

$$\sum_{i=1}^m A_i = I_n.$$

We use the Hilbert–Schmidt inner product and norm

$$\langle U, V \rangle_{\text{HS}} = \text{tr}(U^*V), \quad \|U\|_{\text{HS}}^2 = \text{tr}(U^*U).$$

Set

$$c_m = m(m-1), \quad d_m = 2m-1, \quad S = \sum_i A_i^2, \quad R = \sum_i A_i^3, \quad T = \frac{I_n + c_m R}{d_m}. \quad (1)$$

These definitions are valid on the positive semidefinite boundary. Indeed,

$$S - \frac{1}{m}I_n = \sum_i \left(A_i - \frac{1}{m}I_n\right)^2 \succeq 0, \quad T \succeq \frac{1}{d_m}I_n, \quad (2)$$

so S and T are always positive definite.

Theorem 1.1. *For the matrices above,*

$$\mathrm{tr}(T^{-1}S) \leq n \quad (3)$$

and

$$\det\left(I_n + m(m-1) \sum_i A_i^3\right) \geq (2m-1)^n \det\left(\sum_i A_i^2\right). \quad (4)$$

The constant $(2m-1)^n$ is sharp for each m, n . If every A_i is positive definite and $m \geq 3$, equality in (4) holds if and only if $A_1 = \dots = A_m = I_n/m$. For $m = 1, 2$, both (3) and (4) are equalities for every admissible tuple.

The positive definiteness assumption in the equality classification matters. For example, when $m \geq 2$, the positive semidefinite tuple

$$A_1 = 0, \quad A_2 = \dots = A_m = \frac{1}{m-1} I_n$$

also gives equality, since $S = T = I_n/(m-1)$. We do not classify all equality cases on the boundary.

2 The trace-variational reduction

For a Hermitian matrix X , define

$$\mathcal{K}(X) = \sum_i A_i X A_i, \quad \mathcal{B}(X) = \mathrm{tr}(X T X S) - \|\mathcal{K}(X)\|_{\mathrm{HS}}^2 \quad (5)$$

and

$$\Phi(X) = \mathrm{tr}(I_n - 2XS + X T X S). \quad (6)$$

The expression $\mathrm{tr}(X T X S)$ is real and nonnegative, since it equals $\|T^{1/2} X S^{1/2}\|_{\mathrm{HS}}^2$.

Lemma 2.1. *For every Hermitian X ,*

$$\Phi(X) = \|\mathcal{K}(X) - I_n\|_{\mathrm{HS}}^2 + \mathcal{B}(X), \quad (7)$$

$$\Phi(X) = n - \mathrm{tr}(T^{-1}S) + \|T^{1/2}(X - T^{-1})S^{1/2}\|_{\mathrm{HS}}^2. \quad (8)$$

Consequently, if $\mathcal{B}(X) \geq 0$ for all Hermitian X , then (3) and (4) hold.

Proof. Since $\mathcal{K}(X)$ is Hermitian and $\mathrm{tr} \mathcal{K}(X) = \mathrm{tr}(XS)$, expansion of the squared norm proves (7). Put $Y = X - T^{-1}$. Expansion of $X T X$ gives

$$X T X = Y T Y + 2Y + T^{-1},$$

which proves (8). In particular, $\Phi(T^{-1}) = n - \mathrm{tr}(T^{-1}S)$. Nonnegativity in (7) therefore implies (3).

For the determinant conclusion, the matrix $M = T^{-1/2} S T^{-1/2}$ is positive definite. The arithmetic-geometric mean inequality for its eigenvalues gives

$$\left(\frac{\det S}{\det T}\right)^{1/n} = (\det M)^{1/n} \leq \frac{\mathrm{tr} M}{n} = \frac{\mathrm{tr}(T^{-1}S)}{n} \leq 1. \quad (9)$$

Using (1) yields (4). $\square$

We shall prove the following stronger assertion.

Theorem 2.2. *For every admissible tuple $A_1, \dots, A_m$ and every Hermitian X ,*

$$\mathcal{B}(X) \geq 0. \quad (10)$$

The rest of the proof constructs a Gram representation of $d_m \mathcal{B}(X)$ and proves positivity of its coefficient matrix analytically.

3 A uniform trace–Gram identity

Fix $m \geq 3$ and write $k = m - 1 \geq 2$. For each i , let $J_i = \{1, \dots, m\} \setminus \{i\}$. For $j, l \in J_i$, define

$$E_{ij} = (A_j - A_i)X A_j, \quad F_{ij} = A_j X (A_j - A_i) = E_{ij}^*, \quad (11)$$

$$Z_{i;jl} = (A_j - A_i)X(A_l - A_i) \quad (j \neq l). \quad (12)$$

Let w_i be the column of all E_{ij} , all F_{ij} , and all $Z_{i;jl}$ with ordered pairs $j \neq l$, in any fixed order. It has $2k + k(k - 1) = k(k + 1)$ components. The corresponding abstract component labels are E_j, F_j, Z_{jl} .

Define a real symmetric matrix $\mathcal{H}_k$ on these labels by Table 1; entries in symmetric positions are equal. The table is unchanged by a common relabeling of the k indices. It defines a new parameterized family, distinct from the fixed Gram matrix in Cubesteak’s three-matrix proof.

Table 1: The entries of $\mathcal{H}_k$. Every Z -label has distinct internal indices. The conditions in each block are exhaustive.

Row and column	Condition	Entry
E_j, E_j		$k(k + 5)/2$
E_j, E_l	$j \neq l$	$(k - 3)/2$
F_j, F_j		$k(k + 5)/2$
F_j, F_l	$j \neq l$	$(5k + 1)/2$
E_j, F_j		$-(k + 1)(k + 2)/2$
E_j, F_l	$j \neq l$	$-(k + 1)$
E_j, Z_{lh}	$j = l$	$k/2$
E_j, Z_{lh}	$j \neq l$	$-1/2$
F_j, Z_{lh}	$j = l$	$-(k - 1)/2$
F_j, Z_{lh}	$j = h$	0
F_j, Z_{lh}	$j \notin \{l, h\}$	$1/2$
Z_{jl}, Z_{jl}		$k - 1$
Z_{jl}, Z_{ph}	distinct labels; $j = p$ or $l = h$	$-1/2$
Z_{jl}, Z_{ph}	distinct labels; $j \neq p, l \neq h$	0

For a Hermitian matrix A and a column w of matrices indexed by these labels, put

$$\mathcal{Q}_k(A; w) = \sum_{a,b} (\mathcal{H}_k)_{ab} \operatorname{tr}(w_a^* A w_b). \quad (13)$$

The sum is real, by Hermitian symmetry.

Proposition 3.1. *For every Hermitian X ,*

$$d_m \mathcal{B}(X) = \sum_{i=1}^m \mathcal{Q}_k(A_i; w_i). \quad (14)$$

Equivalently,

$$\sum_i \mathcal{Q}_k(A_i; w_i) = \operatorname{tr}(X^2 S) + m(m - 1) \operatorname{tr}(X R X S) - (2m - 1) \|\mathcal{K}(X)\|_{\text{HS}}^2. \quad (15)$$

This algebraic identity requires only Hermitian symmetry and $\sum_i A_i = I_n$, not positivity of the A_i .

Proof. Expand (13) using (11)–(12) and Table 1. Appendix A gives the full coefficient table, together with the finite index-summation rules used to obtain it. The resulting three nonzero coefficients are 1 , $m(m-1)$, and $1-2m$, on the respective trace words in (15). This proves that equation. Finally, $d_m T = I_n + c_m R$ gives (14). $\square$

4 Analytic positivity of the Gram family

We prove positivity for vectors in an arbitrary complex Hilbert space V , with inner product linear in its second argument. For a vector of components

$$\boldsymbol{\xi} = (\ell_j, r_j, z_{jl})_{1 \leq j, l \leq k, j \neq l},$$

where the ℓ , r , and z entries have the respective E , F , and Z labels, write

$$Q_k(\boldsymbol{\xi}) = \sum_{a,b} (\mathcal{H}_k)_{ab} \langle \xi_a, \xi_b \rangle.$$

All norms and inner products in this section are those of V .

4.1 An orthogonal decomposition of the ordered-pair components

Define

$$\ell_0 = \frac{1}{k} \sum_j \ell_j, \quad r_0 = \frac{1}{k} \sum_j r_j, \quad z_0 = \frac{1}{k(k-1)} \sum_{j \neq l} z_{jl}, \quad \ell_j^\circ = \ell_j - \ell_0, \quad r_j^\circ = r_j - r_0.$$

Let

$$u_j = \sum_{l \neq j} z_{jl} - (k-1)z_0, \quad v_j = \sum_{l \neq j} z_{lj} - (k-1)z_0.$$

Both $\sum_j u_j$ and $\sum_j v_j$ vanish. Put

$$\alpha_j = \begin{cases} (u_j + v_j)/(2(k-2)), & k > 2, \\ 0, & k = 2, \end{cases} \quad \beta_j = \frac{u_j - v_j}{2k}, \quad (16)$$

and

$$\omega_{jl} = z_{jl} - z_0 - \alpha_j - \alpha_l - \beta_j + \beta_l \quad (j \neq l). \quad (17)$$

For $k = 2$, one has $u_j + v_j = 0$, so all formulas below still hold.

We have $\sum_j \alpha_j = \sum_j \beta_j = 0$, and every row sum and column sum of the residual array is zero. In particular,

$$a_j := \sum_{l \neq j} z_{jl} = (k-1)z_0 + (k-2)\alpha_j + k\beta_j, \quad (18)$$

$$b_j := \sum_{l \neq j} z_{lj} = (k-1)z_0 + (k-2)\alpha_j - k\beta_j. \quad (19)$$

The four arrays in

$$z_{jl} = z_0 + (\alpha_j + \alpha_l) + (\beta_j - \beta_l) + \omega_{jl}$$

are mutually orthogonal in the direct-sum Hilbert space indexed by ordered pairs $j \neq l$. Indeed, the zero sums give orthogonality to the constant array. The α -array is symmetric and the β -array is antisymmetric under $j \leftrightarrow l$. Orthogonality to the residual array follows from its zero row and column sums. Expanding the two remaining squared norms yields

$$\sum_{j \neq l} \|z_{jl}\|^2 = k(k-1)\|z_0\|^2 + 2(k-2) \sum_j \|\alpha_j\|^2 + 2k \sum_j \|\beta_j\|^2 + \sum_{j \neq l} \|\omega_{jl}\|^2. \quad (20)$$

4.2 Completion of squares

For a real symmetric 2×2 matrix N , abbreviate

$$\langle (a, b), N(a, b) \rangle = N_{11}\|a\|^2 + 2N_{12} \operatorname{Re}\langle a, b \rangle + N_{22}\|b\|^2.$$

Define

$$N_0 = \frac{1}{4} \begin{pmatrix} 4k^2 + k + 7 & -(6k^2 + 5k + 1) \\ -(6k^2 + 5k + 1) & 12k^2 + k - 1 \end{pmatrix}, \quad (21)$$

$$N_1 = \frac{1}{k+2} \begin{pmatrix} (k+1)(2k+3) & -(k+1)^2 \\ -(k+1)^2 & k^2 + k - 1 \end{pmatrix}. \quad (22)$$

Lemma 4.1. *For every integer $k \geq 2$,*

$$\begin{aligned} Q_k(\xi) &= k(k-1)\|z_0 + \tfrac{1}{2}(\ell_0 - r_0)\|^2 + k\langle (\ell_0, r_0), N_0(\ell_0, r_0) \rangle \\ &\quad + (k-2)(k+2) \sum_j \|\alpha_j + \frac{k+1}{2(k+2)}(\ell_j^\circ - r_j^\circ)\|^2 \\ &\quad + k^2 \sum_j \|\beta_j + \frac{(k+1)\ell_j^\circ - (k-1)r_j^\circ}{2k}\|^2 \\ &\quad + \sum_j \langle (\ell_j^\circ, r_j^\circ), N_1(\ell_j^\circ, r_j^\circ) \rangle + k \sum_{j \neq l} \|\omega_{jl}\|^2. \end{aligned} \quad (23)$$

Consequently, $\mathcal{H}_k \succeq 0$ for $k \geq 2$, and $\mathcal{H}_k \succ 0$ for $k \geq 3$. For $k = 2$, the equality $Q_2(\xi) = 0$ implies

$$\ell_j^\circ = r_j^\circ = 0 \quad (j = 1, 2), \quad 5\ell_0 = 7r_0. \quad (24)$$

Proof. For a transparent verification of (23), put $\ell_\Sigma = \sum_j \ell_j$, $r_\Sigma = \sum_j r_j$, and $z_\Sigma = \sum_{j \neq l} z_{jl}$. Reading off the six blocks of Table 1 gives

$$\begin{aligned} Q_k(\xi) &= \frac{(k+1)(k+3)}{2} \sum_j \|\ell_j\|^2 + \frac{k-3}{2} \|\ell_\Sigma\|^2 \\ &\quad + \frac{(k-1)(k+1)}{2} \sum_j \|r_j\|^2 + \frac{5k+1}{2} \|r_\Sigma\|^2 \\ &\quad - k(k+1) \operatorname{Re} \sum_j \langle \ell_j, r_j \rangle - 2(k+1) \operatorname{Re} \langle \ell_\Sigma, r_\Sigma \rangle \\ &\quad + k \sum_{j \neq l} \|z_{jl}\|^2 - \frac{1}{2} \sum_j (\|a_j\|^2 + \|b_j\|^2) \\ &\quad + \operatorname{Re} \left((k+1) \sum_j \langle \ell_j, a_j \rangle - \langle \ell_\Sigma, z_\Sigma \rangle \right) \\ &\quad + \operatorname{Re} \left(\langle r_\Sigma, z_\Sigma \rangle - k \sum_j \langle r_j, a_j \rangle - \sum_j \langle r_j, b_j \rangle \right). \end{aligned} \quad (25)$$

For instance, the ZZ -block has diagonal $k-1$ and off-diagonal entry $-1/2$ for equal rows or equal columns. Its contribution is exactly the fourth line of (25). The two subsequent lines are the EZ - and FZ -contributions.

Substitute (18)–(20) and $\ell_j = \ell_0 + \ell_j^\circ$, $r_j = r_0 + r_j^\circ$ into (25). Terms between constant and zero-sum components vanish. The coefficients of $\|z_0\|^2$, $\sum_j \|\alpha_j\|^2$, $\sum_j \|\beta_j\|^2$ are, respectively, $k(k-1)$, $(k-2)(k+2)$, k^2 . Completing these three squares leaves the constant block kN_0 , the zero-sum block N_1 , and $k \sum_{j \neq l} \|\omega_{jl}\|^2$, exactly as in (23).

Both upper-left entries in (21)–(22) are positive, and direct determinant expansion gives

$$\det N_0 = \frac{(k-2)(k-1)^2(3k+1)}{4}, \quad \det N_1 = \frac{(k+1)(k^2-2)}{k+2}. \quad (26)$$

Thus $N_0 \succeq 0$ and $N_1 \succ 0$ for $k \geq 2$; both are positive definite for $k \geq 3$. All terms in (23) are nonnegative, proving $\mathcal{H}_k \succeq 0$.

If $k \geq 3$ and $Q_k(\xi) = 0$, the N_0 - and N_1 -terms force $\ell_0 = r_0 = \ell_j^\circ = r_j^\circ = 0$. The other squares then force $z_0 = \alpha_j = \beta_j = \omega_{jl} = 0$. Hence every component is zero, proving strict positivity. For $k = 2$,

$$N_0 = \frac{1}{4} \begin{pmatrix} 25 & -35 \\ -35 & 49 \end{pmatrix}.$$

The N_1 -terms force $\ell_j^\circ = r_j^\circ = 0$; the N_0 -term forces $5\ell_0 = 7r_0$. This proves the necessary conditions (24). $\square$

5 Proofs of the inequalities and the equality statement

Proof of Theorem 2.2. Suppose first that $m \geq 3$. Apply Lemma 4.1 in the Hilbert space $M_n(\mathbb{C})$ with its Hilbert–Schmidt inner product, to the components $A_i^{1/2}(w_i)_a$. Then

$$\mathcal{Q}_k(A_i; w_i) = \sum_{a,b} (\mathcal{H}_k)_{ab} \langle A_i^{1/2}(w_i)_a, A_i^{1/2}(w_i)_b \rangle_{\text{HS}} \geq 0.$$

Proposition 3.1 and $d_m > 0$ prove $\mathcal{B}(X) \geq 0$.

For $m = 1$, $A_1 = S = T = I_n$, so $\mathcal{B}(X) = 0$. For $m = 2$, write $A_2 = I_n - A_1$. Direct expansion gives

$$I_n + 2(A_1^3 + (I_n - A_1)^3) = 3(A_1^2 + (I_n - A_1)^2),$$

and hence $T = S$. The two matrices A_i are simultaneously diagonalizable. In a common orthonormal eigenbasis write their diagonal entries as a_{ip} , and put

$$s_p = \sum_{i=1}^2 a_{ip}^2, \quad h_{pq} = \sum_{i=1}^2 a_{ip} a_{iq}.$$

For Hermitian $X = (x_{pq})$,

$$\text{tr}(XSXS) = \sum_{p,q} s_p s_q |x_{pq}|^2, \quad \|\mathcal{K}(X)\|_{\text{HS}}^2 = \sum_{p,q} h_{pq}^2 |x_{pq}|^2.$$

Cauchy–Schwarz gives $h_{pq}^2 \leq s_p s_q$. Therefore $\mathcal{B}(X) \geq 0$ also for $m = 2$. $\square$

Proof of Theorem 1.1. The two inequalities follow from Theorem 2.2 and Lemma 2.1. The preceding proof also shows $S = T$ for $m = 1, 2$, giving the asserted identities in these cases. For any m , the choice $A_i = I_n/m$ gives

$$S = T = \frac{1}{m} I_n,$$

so equality holds and the constant is sharp.

It remains to prove uniqueness of equality in the positive definite domain when $m \geq 3$. Assume $\det S = \det T$. Both inequalities in (9) must be equalities. In particular, $\text{tr}(T^{-1}S) = n$. Set $X = T^{-1} \succ 0$. Equations (7) and (14) give

$$0 = \Phi(X) = \|\mathcal{K}(X) - I_n\|_{\text{HS}}^2 + \frac{1}{d_m} \sum_i \mathcal{Q}_k(A_i; w_i).$$

Every summand is nonnegative; hence $\mathcal{Q}_k(A_i; w_i) = 0$ for each i .

If $m \geq 4$, then $k \geq 3$ and $\mathcal{H}_k \succ 0$. It follows that $A_i^{1/2} E_{ij} = 0$ for every $j \neq i$. Since A_i is positive definite, $E_{ij} = 0$; thus

$$(A_j - A_i)X A_j = 0.$$

Both X and A_j are invertible, so $A_j = A_i$. All matrices are therefore I_n/m .

If $m = 3$, then $k = 2$. Apply (24) to $\ell_j = A_i^{1/2} E_{ij}$ and $r_j = A_i^{1/2} F_{ij}$, with the two elements of J_i identified with 1, 2. Canceling the invertible factor $A_i^{1/2}$, the two E_{ij} 's have a common value E_i^0 , the two F_{ij} 's have a common value F_i^0 , and

$$5E_i^0 = 7F_i^0.$$

By (11), $F_i^0 = (E_i^0)^*$. Taking adjoints also gives $5F_i^0 = 7E_i^0$, whence $E_i^0 = F_i^0 = 0$. The same invertibility argument yields $A_j = A_i$, completing the equality classification. $\square$

A Exact expansion of the trace–Gram identity

This appendix provides the algebraic details of Proposition 3.1. All reductions use only real trace identities, finite sums, and $\sum_i A_i = I_n$.

A.1 Trace words and the complete coefficient table

For a word W in the symbols A, B, X , define $\Theta(W)$ as follows. Replace each distinct symbol among A, B occurring in W by an independently summed matrix A_i , leave X fixed, and take the real part of the trace. For example,

$$\Theta(\text{AAAXBXB}) = \sum_{i,j} \text{Re tr}(A_i^3 X A_j^2 X) = \text{tr}(X R X S), \quad \Theta(\text{AAXX}) = \text{tr}(X^2 S).$$

Different summation labels are independent and may have equal values. Cyclic rotation, reversal, and consistent renaming of the summed symbols leave Θ unchanged. Reversal is valid because every letter represents a Hermitian matrix and the real trace is taken. Also,

$$\Theta(\text{ABXBAX}) = \sum_{i,j} \text{Re tr}(A_i A_j X A_j A_i X) = \|\mathcal{K}(X)\|_{\text{HS}}^2.$$

Expand $\sum_i \mathcal{Q}_k(A_i; w_i)$ and separate its six contributions according to component types. The columns EF, EZ, FZ each include both orders of the corresponding cross terms. Table 2 lists every nonzero word coefficient after setting $k = m - 1$. A blank entry means zero; all words not listed have zero coefficient. The entries are polynomials in m , not values at selected integers.

Adding across each row leaves only

$$m(m-1)\Theta(\text{AAAXBXB}) + \Theta(\text{AAXX}) + (1-2m)\Theta(\text{ABXBAX}),$$

which is (15).

A.2 Finite summation rules underlying the table

For completeness, we specify an exact procedure for obtaining all entries of Table 2, including the treatment of distinct indices. Before expansion, a term has one distinguished index i and at most four other index positions. Separate these terms according to the equality pattern of the latter positions, keeping i distinct from all of them. Each pattern is counted once and is summed over all assignments of pairwise distinct indices to its distinct labels. The Gram coefficient is determined by Table 1 for that pattern.

Table 2: Complete trace-word expansion of the six Gram blocks.

Word	EE	EF	EZ	FF	FZ	ZZ
AAAXBBX	$\frac{m^2}{2} + m$		$-m$			$\frac{m^2}{2} - m$
AAAXX	$\frac{m}{2} - 2$		1			$1 - \frac{m}{2}$
AABXABX		$m - m^2$	$m^2 - m$			
AABXBBX	$-m^2 - 2m$	$m^2 - m$	$2m$			m
AABXBX	$4 - m$	$2m$	$-2 - m$			-2
AAXAAX	$\frac{m^2}{2} + m$	$m - m^2$	$-m$	$\frac{m^2}{2} - m$	m	$-m$
AAXAX		$-2m$	1	$-2 + \frac{5m}{2}$	-2	$3 - \frac{m}{2}$
AAXBABX				$\frac{m^2}{2} - m$	$2m - m^2$	$\frac{m^2}{2} - m$
AAXX			1			
ABAXABX		$m^2 - m$	$m - m^2$	$2m - m^2$	$m^2 - 3m$	m
ABAXBX					$m - 1$	$1 - m$
ABXABX					1	-1
ABXBAX	$\frac{m}{2} - 2$		m	$2 - \frac{5m}{2}$	$2 - m$	-1
AXAX			-1			1

The following finite inclusion–exclusion identity changes such distinct-index sums into unrestricted sums. For any function f of r indices in $\{1, \dots, m\}$,

$$\sum_{\substack{i_1, \dots, i_r \\ \text{pairwise distinct}}} f(i_1, \dots, i_r) = \sum_{\pi \in \Pi_r} \left[\prod_{C \in \pi} (-1)^{|C|-1} (|C| - 1)! \right] \sum_{\substack{i_1, \dots, i_r \\ i_a = i_b \text{ if } a, b \in C \in \pi}} f(i_1, \dots, i_r). \quad (27)$$

Here Π_r denotes the set of set partitions of $\{1, \dots, r\}$; indices belonging to different blocks in the inner sum remain independent and are allowed to coincide.

To prove (27), fix an assignment of all the indices and let ρ be its equality partition. Its coefficient on the right is the sum of the displayed partition weights over π refining ρ . This factors over the blocks of ρ . For a block of size s , the factor is the sum of the signs of all permutations of s objects: a partition specifies their cycle supports, and a block of size t has $(t - 1)!$ cyclic orderings, each contributing sign $(-1)^{t-1}$. The sign sum is 1 for $s = 1$ and 0 for $s \geq 2$, since composition with a fixed transposition pairs opposite signs. Thus the total coefficient is 1 exactly for pairwise distinct indices and 0 otherwise, as required.

After applying (27), expand each E, F, Z into its two or four monomials. Every resulting trace word initially has five A -factors and two X -factors, because each entry of w_i has two A -factors and one X -factor and (13) inserts one additional A_i . An unrestricted summed label appearing exactly once can be removed: summing its single matrix factor replaces it by $\sum_i A_i = I_n$. A summed label absent from the word contributes a scalar factor m . Consequently at most two matrix labels remain, as each remaining label occurs at least twice among at most five matrix factors.

Finally identify words under cyclic rotation, reversal, and renaming of summed labels, choosing the lexicographically least representative after renaming labels by order of first occurrence. Expanding with these rules and collecting the six component types yields precisely Table 2. All

operations involve at most five index labels, so (27) uses only $r \leq 5$. This argument is valid also when $r > m$, in which case the distinct-index sum is empty and the right side cancels to zero. Thus the table and the Gram identity hold for every integer $m \geq 3$ without an interpolation or limiting argument.

B The accompanying exact certificate

The sole accompanying program is

`stage6_round2_identity_certificate.py.`

It independently checks (14) by implementing the finite expansion rules of Appendix A. Polynomial coefficients are stored exactly using rational numbers; the parameter m is indeterminate. Four auxiliary labels suffice to enumerate every equality pattern of a pair of component labels; the distinguished index gives at most five labels in (27). The program collects the coefficients of the normalized trace words and checks that their sum is exactly

$$\Theta(\text{AAXX}) + (m^2 - m)\Theta(\text{AAAXBX}) + (1 - 2m)\Theta(\text{ABXBAX}).$$

It uses only the Python standard library. Running

```
python3 stage6_round2_identity_certificate.py
```

produces

```
PASS: symbolic identity for indeterminate m
```

The certificate performs no matrix sampling, floating-point optimization, or extrapolation from finitely many values of m . It is an optional verification of the algebraic identity. Positivity of the Gram family, the trace reduction, and the equality classification are proved analytically in the paper.

References

- [1] Cubesteak, *Answer concerning the determinant inequality* $\det(I_n + 6(A^3 + B^3 + C^3)) \geq 5^n \det(A^2 + B^2 + C^2)$, Mathematics Stack Exchange, answer no. 5147902. <https://math.stackexchange.com/a/5147902>.